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Prof. Guangtao Zhai
Editor-in-Chief
Displays
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Dear Professor Zhai,

We transfer the manuscript “**Pseudo-3D Heightfield Reconstruction from Endoscopic Video via Normal-Guided Multi-View Optimization and Metric Scale Gauge Injection**” to *Displays* as a Research Article. The revised manuscript treats the heightfield as a surface that can be displayed and, separately, as a surface whose measurement error is reported.

Why Displays. The useful output is a renderable mesh for intra-operative overlay and measurement, not a disparity map alone. A collapsed cone or planar slab cannot be visualized as a surgical surface and cannot be used to measure physical lengths. This addresses a fundamental display problem: visual signal processing, an un-collapsed 3D surface, and explicit metric scale recovery.

What changed for this transfer.

- **Normal-guided branch disambiguation:** We resolve the mathematical two-fold ambiguity in small-baseline planar homography decomposition using an operative physical normal prior $\mathbf{n}_{hf}$ derived from the heightfield mesh, vetoing the physically inconsistent branch that cheirality heuristics mis-select on a quarter of frame pairs, without external sensors.
- **Multi-stride co-visibility Bundle Adjustment (BA):** We replace sequential dead-reckoning pose chaining with a multi-stride co-visibility graph across non-adjacent keyframes, optimizing SE(3) Lie-algebra camera poses and triangulated 3D points jointly under a robust Huber loss.
- **Metric scale gauge injection:** We inject a physical metric scale from calibration landmarks ($s_{sq} = 3.0\text{ mm}$) to resolve the fundamental scale ambiguity of monocular reconstruction. On our newly constructed 17-sequence surgical benchmark (13 real, 4 simulation), the pattern-free multi-view BA attains a mean ATE of 35.52 mm against ORB-SfM’s 43.72 mm and wins the smoke-filled cavity s10097 (24.86 mm vs. VGGT-1B’s 37.25 mm); when the calibration gauge is fully visible, the gauge-referenced trajectory reaches the pattern-localisation floor of 4.49 mm median ATE (vs. VGGT-1B’s 16.86 mm).

- **Stereo validation and baseline gauge:** On public EndoNeRF, the calibrated stereo variant achieves an overall AbsRel of 0.0285, outperforming FoundationStereo (0.0321) and Stereo SGM (0.0323). On the chessboard benchmark, sparse gauge triangulation yields a held-out median MAE of 1.45 mm, outperforming COLMAP (3.41 mm) and VGGT (4.65 mm).

Declarations. This manuscript is not under consideration elsewhere. All authors have approved the submission. The study uses public datasets and in-house bench-top phantom recordings; it does not involve new human-subject or animal experiments. The authors have no competing interests. Code, evaluation scripts, and result files are at <https://github.com/liranyang123456-commits/re3d-heightfield>. Full in-house sequences remain available from the corresponding author.

We thank you for considering this work.

Sincerely,

Ranyang Li, et al.